Poisson Distribution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Conditions for a Poisson distribution**

State the key conditions required for a random variable to be modeled by a Poisson distribution.

Events occur independently of one another; events occur at a constant average rate over the given interval of time or space; two events cannot occur at exactly the same instant; the number of events in non-overlapping intervals is independent.

2 **[Foundational]** • **Target: Conditions for a Poisson distribution**

Explain why the number of texts a person receives in a randomly chosen hour might reasonably be modeled by a Poisson distribution, but the number of texts received right around a pre-arranged deadline might not.

Texts received during a typical hour occur independently and at a roughly constant average rate, satisfying the Poisson conditions. Texts clustered around a pre-arranged deadline (e.g. everyone replying just before a cutoff) violate the constant-rate and independence assumptions, since there is a systematic surge rather than random, uniform occurrence.

3 **[Foundational]** • **Target: Computing Poisson probabilities**

A shop receives an average of 3 customers per minute, modeled by $X \sim \text{Po}(3)$. Find $P(X = 5)$, to three decimal places.

$P(X = 5) = \frac{e^{-3}3^5}{5!} = \frac{(0.049787)(243)}{120} \approx 0.101.$

4 **[Foundational]** • **Target: Computing Poisson probabilities**

Using the same shop, find $P(X = 1)$, to three decimal places.

$P(X = 1) = 3e^{-3} \approx 3(0.049787) \approx 0.149.$

5 **[Foundational]** • **Target: Computing Poisson probabilities**

Using the same shop, find $P(X \leq 2)$.

$P(X = 0) = e^{-3} \approx 0.0498$. $P(X = 1) \approx 0.1494$. $P(X = 2) = \frac{9e^{-3}}{2} \approx 0.2240$. Sum: $P(X \leq 2) \approx 0.423$.

6 **[Foundational]** • **Target: Mean and variance of a Poisson distribution**

State the mean and variance of $X \sim \text{Po}(\lambda)$, and find the standard deviation when $\lambda = 9$.

$E(X) = \lambda$ and $\text{Var}(X) = \lambda$ — both equal λ, unlike most other distributions where mean and variance are independent parameters. When $\lambda = 9$: standard deviation $= \sqrt{9} = 3$.

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7 [Foundational] • Target: Mean and variance of a Poisson distribution

A random variable $X \sim \text{Po}(7)$. State $E(X)$, $\text{Var}(X)$, and the standard deviation, to three decimal places.

$E(X) = 7$, $\text{Var}(X) = 7$, standard deviation $= \sqrt{7} \approx 2.646$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums of independent Poisson variables

Two independent Poisson processes have rates $\lambda_1 = 4$ and $\lambda_2 = 6$ per hour. State the distribution of the total number of events, and find its mean and variance.

The sum of independent Poisson variables is itself Poisson: total $\sim \text{Po}(4 + 6) = \text{Po}(10)$. Mean $= 10$; variance $= 10$.

9 [Intermediate] • Target: Sums of independent Poisson variables

Three independent Poisson processes have rates $\lambda_1 = 2$, $\lambda_2 = 4$, $\lambda_3 = 5$ per hour. Find the mean and variance of the total number of events across all three processes in one hour.

Total $\sim \text{Po}(2 + 4 + 5) = \text{Po}(11)$. Mean $= 11$; variance $= 11$.

10 [Intermediate] • Target: Poisson as an approximation to the binomial

Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the value of λ to use.

The approximation is appropriate when n is large and p is small (a common guideline: $n > 50$ and $np < 5$), so events are rare relative to the number of trials. Then X is approximately $\text{Po}(\lambda)$ with $\lambda = np$.

11 [Intermediate] • Target: Poisson as an approximation to the binomial

A factory produces 2000 items, each independently defective with probability 0.0025. Using the Poisson approximation, find the probability that exactly 2 items are defective, to three decimal places.

$\lambda = np = 2000(0.0025) = 5$. $P(X = 2) = \frac{e^{-5}5^2}{2!} = \frac{(0.006738)(25)}{2} \approx 0.084$.

12 [Intermediate] • Target: Poisson as an approximation to the binomial

A survey mails 5000 questionnaires, each independently containing a printing error with probability 0.0006. Using the Poisson approximation, find the probability that no questionnaires contain an error, to four decimal places.

$\lambda = np = 5000(0.0006) = 3$. $P(X = 0) = e^{-3} \approx 0.0498$.

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13 [Intermediate] • Target: Using cumulative Poisson probabilities

A call center receives calls at $\lambda = 6$ per 10-minute interval. Find the probability of receiving more than 3 calls, given $P(X = 0) \approx 0.00248$, $P(X = 1) \approx 0.01487$, $P(X = 2) \approx 0.04462$, $P(X = 3) \approx 0.08924$.

$P(X > 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)] = 1 - 0.15121 \approx 0.849$.

14 [Intermediate] • Target: Using cumulative Poisson probabilities

A website receives visitors at $\lambda = 4$ per minute. Find $P(1 \le X \le 3)$, given $P(X = 0) \approx 0.0183$, $P(X = 1) \approx 0.0733$, $P(X = 2) \approx 0.1465$, $P(X = 3) \approx 0.1954$.

$P(1 \le X \le 3) = P(X = 1) + P(X = 2) + P(X = 3) = 0.0733 + 0.1465 + 0.1954 \approx 0.415$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 13.01 Continuous Random Variables

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.0498$. Find λ, to the nearest whole number.

$e^{-\lambda} = 0.0498 \Rightarrow -\lambda = \ln(0.0498) \approx -3.00 \Rightarrow \lambda \approx 3$.

16 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 14.01 Sampling & Estimation

A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 2. Find λ, and then find $P(X = \lambda)$, to three decimal places.

Since variance $= \lambda$ and standard deviation $= \sqrt{\lambda} = 2$, $\lambda = 4$. $P(X = 4) = \frac{e^{-4}4^4}{4!} = \frac{(0.018316)(256)}{24} \approx 0.195$.

17 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 15.01 Hypothesis Testing

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 1) = 0.1494$, where λ is a whole number less than 5. Using $P(X = 1) = \lambda e^{-\lambda}$, determine λ by trial.

Testing $\lambda = 3$: $3e^{-3} = 3(0.049787) \approx 0.1494$ ✓, matching the given probability. So $\lambda = 3$.

18 [Challenging] • Target: Poisson processes over different intervals and conditional probability • Cross-Lesson: 2.01 Exponentials & Logarithms

Machine faults occur at an average rate of $\lambda = 2$ per day. Find the probability that exactly 2 faults occur today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.1353$ and $P(X = 2) \approx 0.2707$.

$P(X \ge 1) = 1 - P(X = 0) = 1 - 0.1353 = 0.8647$. Conditional probability:
$P(X = 2 \mid X \ge 1) = \frac{0.2707}{0.8647} \approx 0.313$.

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19 [Challenging] • Target: Poisson processes over different intervals and conditional probability • Cross-Lesson: 3.01 Advanced Trigonometry

A radioactive source emits particles at a rate of $\lambda = 1.2$ per second, modeled by a Poisson distribution. Find $P(X = 2)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

For a 3-second interval, $\lambda = 1.2 \times 3 = 3.6$. $P(X = 2) = \frac{e^{-3.6}(3.6)^2}{2!} = \frac{(0.027324)(12.96)}{2} \approx 0.177$.

20 [Challenging] • Target: A well-designed word problem: quality control at a bottling plant • Cross-Lesson: 4.01 Differentiation

This question has four parts. Faulty caps occur at random in bottling cases with an average rate of $\lambda = 1.5$ per case. (a) Find $P(X = 0)$ and $P(X = 1)$ for a randomly selected case, to four decimal places. (b) Find the probability that a case has at least 2 faulty caps. (c) The plant inspects 40 cases per day. Using the fact that a sum of independent Poisson variables is itself Poisson, state the distribution of the total number of faulty caps found across all 40 cases in a day, and find its mean. (d) Since this total has a large λ, it can be approximated by $N(\lambda, \lambda)$. Using a continuity correction and $\Phi(1.36) \approx 0.9131$, estimate the probability that more than 70 faulty caps are found in a day.

(a) $P(X = 0) = e^{-1.5} \approx 0.2231$. $P(X = 1) = 1.5e^{-1.5} \approx 0.3347$. (b) $P(X \geq 2) = 1 - 0.2231 - 0.3347 \approx 0.4422$. (c) Total $\sim \text{Po}(40 \times 1.5) = \text{Po}(60)$, with mean 60. (d) Using $N(60, 60)$ with $\sigma = \sqrt{60} \approx 7.746$, and a continuity correction for "more than 70" (i.e. ≥ 71): $z = \frac{70.5 - 60}{7.746} \approx 1.36$. $P(\text{total} > 70) \approx 1 - \Phi(1.36) \approx 1 - 0.9131 \approx 0.087$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Poisson Probability Formula & Conditions

Which of the following is NOT an essential assumption for modeling a random variable with a Poisson distribution $\text{Po}(\lambda)$?

a. Events occur at a variable rate over time ✓

b. Events occur singly and independently

c. The average rate of occurrence λ is constant

d. Two events cannot occur at the exact same instant

In a Poisson process, events must occur at a constant average rate, independently, at random, and singly.

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22 [Foundational] • Target: Poisson Probability Formula & Conditions

The random variable $X \sim \text{Po}(3.2)$. Find $P(X = 2)$ correct to 4 decimal places.

a. 0.2087 ✓

b. 0.2226

c. 0.1782

d. 0.1450

$P(X = 2) = \frac{e^{-3.2}(3.2)^2}{2!} = \frac{0.0407622 \times 10.24}{2} = 0.2087.$

23 [Foundational] • Target: Poisson Probability Formula & Conditions

If $X \sim \text{Po}(\lambda)$ and $P(X = 0) = 0.05$, find the value of λ.

a. $\lambda = -\ln(0.05) \approx 2.996$ ✓

b. $\lambda = 0.05$

c. $\lambda = e^{0.05} \approx 1.051$

d. $\lambda = 20$

$P(X = 0) = e^{-\lambda} = 0.05 \Rightarrow -\lambda = \ln(0.05) \Rightarrow \lambda = -\ln(0.05) \approx 2.996.$

24 [Foundational] • Target: Mean, Variance & Linear Combinations

What is the theoretical relationship between the mean and variance of a Poisson distribution $X \sim \text{Po}(\lambda)$?

a. **Mean** $= \text{Variance} = \lambda$ ✓

b. Mean $= \lambda, \text{Variance} = \lambda^2$

c. Mean $= \lambda, \text{Variance} = \sqrt{\lambda}$

d. Mean $= \lambda, \text{Variance} = \lambda(1 - p)$

A defining property of the Poisson distribution is that its mean and variance are equal: $E(X) = \text{Var}(X) = \lambda$.

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25 [Foundational] • Target: Mean, Variance & Linear Combinations

Independent random variables $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(3)$ represent customer arrivals. What distribution does $X + Y$ follow?

a. $\text{Po}(5)$ ✓

b. $\text{Po}(6)$

c. $\text{Po}(2.5)$

d. $N(5, 5)$

The sum of independent Poisson variables is also Poisson: $X + Y \sim \text{Po}(\lambda_X + \lambda_Y) = \text{Po}(2 + 3) = \text{Po}(5)$.

26 [Foundational] • Target: Mean, Variance & Linear Combinations

Faults on a fiber optic cable occur at an average rate of 0.4 per kilometer. What is the probability of finding exactly 1 fault in a 5-kilometer section?

a. 0.2707 ✓

b. 0.4000

c. 0.1353

d. 0.3250

Rate for 5 km is $\lambda = 0.4 \times 5 = 2.0$. Then $P(X = 1) = \frac{e^{-2}(2)^1}{1!} = 2e^{-2} = 2(0.135335) \approx 0.2707$.

27 [Foundational] • Target: Poisson Approximation to the Binomial

Under which conditions can a binomial distribution $B(n, p)$ be well approximated by a Poisson distribution?

a. $n > 50$ **and** $np < 5$ ✓

b. $n > 30$ and $p = 0.5$

c. $np > 5$ and $n(1 - p) > 5$

d. $n < 20$ and $p > 0.1$

The Poisson approximation to the binomial is appropriate when the number of trials n is large ($n > 50$) and the probability of success p is small ($np < 5$), with parameter $\lambda = np$.

28 [Foundational] • Target: Poisson Approximation to the Binomial

A book of 400 pages contains 200 typographical errors distributed at random. Use a Poisson approximation to find the probability that a randomly chosen page contains no errors.

a. $e^{-0.5} \approx 0.6065$ ✓

b. 0.5000

c. 0.3679

d. 0.7500

Average errors per page $\lambda = 200/400 = 0.5$. $P(X = 0) = e^{-0.5} \approx 0.6065$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Poisson Approximation to the Binomial**

A factory produces screws, of which 1% are defective. In a random sample of 250 screws, what is the Poisson parameter λ to model defective screws?

a. $\lambda = 2.5$ ✓

b. $\lambda = 0.01$

c. $\lambda = 25$

d. $\lambda = 5$

$\lambda = np = 250 \times 0.01 = 2.5$.

30 **[Intermediate]** • **Target: Normal Approximation to the Poisson Distribution**

Under what condition can a Poisson distribution $X \sim \text{Po}(\lambda)$ be approximated by a normal distribution?

a. $\lambda > 15$ ✓

b. $\lambda < 5$

c. λ is an integer

d. $\lambda > 100$ only

When $\lambda > 15$, the distribution is sufficiently symmetric to be approximated by $N(\lambda, \lambda)$.

31 **[Intermediate]** • **Target: Normal Approximation to the Poisson Distribution**

If $X \sim \text{Po}(36)$, which normal distribution provides the best approximation?

a. $N(36, 36)$ **with mean 36 and variance 36** ✓

b. $N(36, 6)$

c. $N(0, 1)$

d. $N(36, 18)$

Since mean and variance of $\text{Po}(\lambda)$ are both λ, the approximating normal distribution is $N(\lambda, \lambda) = N(36, 36)$.

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32 [Intermediate] • Target: Normal Approximation to the Poisson Distribution

For $X \sim \text{Po}(36)$, use the normal approximation with continuity correction to state the calculation for $P(X \leq 40)$.

a. $P(Y < 40.5)$ **where** $Y \sim N(36, 36)$ ✓

b. $P(Y < 40)$ where $Y \sim N(36, 36)$

c. $P(Y < 39.5)$ where $Y \sim N(36, 36)$

d. $P(Y < 41)$ where $Y \sim N(36, 6)$

For discrete $X \leq 40$, the continuity correction extends to the upper boundary 40.5: $P(Y < 40.5)$.

33 [Intermediate] • Target: Normal Approximation to the Poisson Distribution

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

a. **0.980** ✓

b. 0.900

c. 0.810

d. 0.990

1. **Margin of error for each confidence interval:** For a symmetric 90% confidence interval with known σ, the critical value is $z = 1.645$. Each interval has half-width: $E = 1.645\frac{\sigma}{\sqrt{n}}$ The two intervals are $I_1 = [\bar{X}_1 - E, \bar{X}_1 + E]$ and $I_2 = [\bar{X}_2 - E, \bar{X}_2 + E]$. 2. **Condition for interval overlap:** Two intervals of identical half-width E overlap if and only if the distance between their centres does not exceed the sum of their half-widths.

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34 [Intermediate] • Target: Conditions for a Poisson distribution

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X + Y = 3)$.

a. $\frac{19}{27}$ ✓

b. $\frac{8}{27}$

c. $\frac{7}{8}$

d. $\frac{2}{3}$

Let $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(2\lambda)$ be independent Poisson random variables. The sum $T = X + Y$ follows a Poisson distribution with parameter $\lambda + 2\lambda = 3\lambda$. The conditional distribution of X given $X + Y = n$ is binomial $\mathrm{B}\left(n, \frac{\lambda}{\lambda + 2\lambda}\right) = \mathrm{B}\left(n, \frac{1}{3}\right)$. Here, $n = 3$, so $X \mid (X + Y = 3) \sim \mathrm{B}\left(3, \frac{1}{3}\right)$. We want to find.

35 [Intermediate] • Target: Computing Poisson probabilities

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests: $H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$ The two-stage procedure operates as follows: - **Stage 1:** A $1-\text{metre}$ sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2. - **Stage 2:** A second, independent $1-\text{metre}$ sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted. Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

a. $\mathrm{e}^{-2} + 16\mathrm{e}^{-4} \approx 0.428$ ✓

b. $16\mathrm{e}^{-4} \approx 0.293$

c. $4\mathrm{e}^{-2} \approx 0.541$

d. $1 - \mathrm{e}^{-1} - \frac{7}{2}\mathrm{e}^{-2} \approx 0.158$

A Type II error occurs when H_0 is accepted despite the true parameter value being $\lambda = 2$ (i.e. H_1 is true). Under $\lambda = 2$, the independent counts $X_1, X_2 \sim \mathrm{Po}(2)$. The probability mass function is: $P(X = k) = \frac{2^k \mathrm{e}^{-2}}{k!}$ Specifically: $P(X = 0) = \mathrm{e}^{-2}, \quad P(X = 1) = 2\mathrm{e}^{-2}, \quad P(X = 2) = 2\mathrm{e}^{-2}$ H_0 can be accepted in two mutually exclusive ways: 1. **In Stage 1:** H_0 is accepted immediately if $X_1 = 0$.

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36 [Intermediate] • Target: Mean and variance of a Poisson distribution

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. **44** ✓

b. 43

c. 11

d. 22

We solve the problem in two stages: **Step 1: Determine σ** Under H_0, the distribution of the sample mean is $\bar{X} \sim N\left(20.0, \frac{\sigma^2}{25}\right)$. The critical value for a upper one-tailed test at the 5% level corresponds to $z = 1.645$: $\frac{20.658 - 20.0}{\frac{\sigma}{\sqrt{25}}} = 1.645$ $\frac{0.658}{\frac{\sigma}{5}} = 1.645 \Rightarrow \frac{3.29}{\sigma} = 1.645 \Rightarrow \sigma = \frac{3.29}{1.645} = 2.0$ mm **Step 2: Determine the minimum sample size m** For.

37 [Intermediate] • Target: Sums of independent Poisson variables

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$**; reject** H_0 **because** $-2.0 < -1.645$ ✓

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

1. Under the null hypothesis $H_0 : \mu = 25.0$, the sample mean $\bar{X}$ of a random sample of size $n = 16$ from $\text{N}(\mu, \sigma^2)$ has standard error: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{16}} = \frac{0.8}{4} = 0.2$ 2. The standardized test statistic z is: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{24.6 - 25.0}{0.2} = \frac{-0.4}{0.2} = -2.0$ 3. Since this is a lower-tail test at the 5% level with critical value $z_{\text{crit}} = -1.645$, the critical (rejec.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Poisson as an approximation to the binomial** • **Cross-Lesson: 2.01 Exponentials & Logarithms**

$X \sim Po(13)$ and $Y \sim Po(13)$ are independent. Find the parameter of the distribution of $X + Y$.

a. **26** ✓

b. 27

c. 169

d. 0

Since X and Y are independent Poisson variables, $X + Y \sim Po(13 + 13) = Po(26)$.

39 **[Challenging]** • **Target: Using cumulative Poisson probabilities** • **Cross-Lesson: 3.01 Advanced Trigonometry**

$X \sim Po(15)$ and $Y \sim Po(11)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 27

b. 165

c. 4

d. **26** ✓

Since X and Y are independent Poisson variables, $X + Y \sim Po(15 + 11) = Po(26)$.

40 **[Challenging]** • **Target: Finding an unknown parameter** • **Cross-Lesson: 4.01 Differentiation**

$X \sim Po(18)$ and $Y \sim Po(19)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 342

b. 38

c. 1

d. **37** ✓

Since X and Y are independent Poisson variables, $X + Y \sim Po(18 + 19) = Po(37)$.

41 **[Challenging]** • **Target: Poisson processes over different intervals and conditional probability** • **Cross-Lesson: 5.01 Integration**

$X \sim Po(19)$ and $Y \sim Po(25)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 475

b. 45

c. **44** ✓

d. 6

Since X and Y are independent Poisson variables, $X + Y \sim Po(19 + 25) = Po(44)$.

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42 [Challenging] • Target: A well-designed word problem: quality control at a bottling plant • Cross-Lesson: 6.01 Sequences & Series

$X \sim Po(18)$ and $Y \sim Po(22)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 396

b. 4

c. **40** ✓

d. 41

Since X and Y are independent Poisson variables, $X + Y \sim Po(18 + 22) = Po(40)$.

43 [Challenging] • Target: Poisson Probability Formula & Conditions • Cross-Lesson: 7.01 Vectors

$X \sim Po(2)$ and $Y \sim Po(14)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 17

b. **16** ✓

c. 28

d. 12

Since X and Y are independent Poisson variables, $X + Y \sim Po(2 + 14) = Po(16)$.

44 [Challenging] • Target: Mean, Variance & Linear Combinations • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

$X \sim Po(7)$ and $Y \sim Po(14)$ are independent. Find the parameter of the distribution of $X + Y$.

a. **21** ✓

b. 22

c. 7

d. 98

Since X and Y are independent Poisson variables, $X + Y \sim Po(7 + 14) = Po(21)$.

45 [Challenging] • Target: Poisson Approximation to the Binomial • Cross-Lesson: 9.01 Complex Numbers

$X \sim Po(7)$ and $Y \sim Po(15)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 105

b. **22** ✓

c. 23

d. 8

Since X and Y are independent Poisson variables, $X + Y \sim Po(7 + 15) = Po(22)$.

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